EART11300

Physical properties of minerals

Video 4: Colour and hardness of minerals

Reading:
Section on colour in the online mineralogy textbook:
https://opengeology.org/Mineralogy/3-properties/#343_Color
Section on hardness in the online mineralogy textbook:
https://opengeology.org/Mineralogy/3-properties/#353_Hardness

Colour of minerals

- The impression of colour in minerals results from the preferential absorption of certain wavelengths from white light, so that the complementary colour is seen. Although colour is the mineral property which makes the most immediate impact, its causes are complex, and it must be used as a diagnostic property with some care.
- Some minerals are intrinsically coloured because of their chemical composition (this applies particularly to minerals containing essential Ti, V, Cr, Mn, Fe, Ni, Co and Cu).
- Most other minerals are essentially colourless, but may occur as distinctly coloured crystals in the natural state due to impurities.

Olivine is green because it contains Fe.

Polished gem-quality olivine crystals, showing characteristic colour

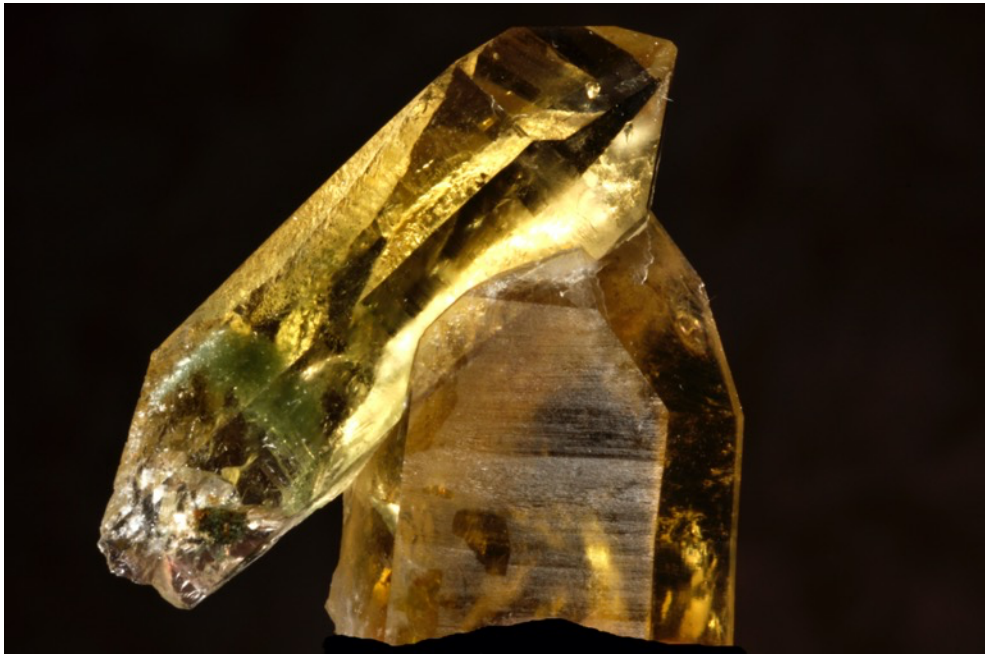


Quartz is intrinsically colourless, but may contain impurities giving it colours such as in amethyst (Fe^{2+}) and citrine (Fe^{3+}).

Images from Wikipedia







Links to videos about colour in minerals:

- Garnet
- Amethyst (quartz)

Hardness

- Defined as resistance to scratching.
- Useful diagnostic property.
- Note that hardness can vary with direction in a mineral because it depends on bond density.
- Usually measured by reference to a set of standard minerals – Mohs' Scale of Hardness (1812). This is a qualitative test: each mineral is capable of scratching the minerals with lower hardness.

1. talc
2. gypsum
3. calcite
4. fluorite
5. apatite
6. orthoclase
7. quartz
8. topaz
9. corundum
10. diamond